Indian Statistical Institute, Bangalore
B. Math.
First Year, Second Semester

Linear Algebra II — Instructor: B V Rajarama Bhat
Home Assignment V — Due Date : April 19, 2026

Remark: Standard inner product is considered on $\mathbb{R}^n$ and $\mathbb{C}^n$ unless some other inner product is explicitly mentioned and $\{e_1, \ldots, e_n\}$ would denote the standard orthonormal basis. All projections considered are orthogonal projections.

(1) (i) Let $\{a_1, a_2, \ldots, a_k\}$ be the set of all distinct eigenvalues of a matrix A. Consider the polynomial $q(x) = (x - a_1)(x - a_2)\cdots(x - a_k)$. Show that A is diagonalizable if and only if $q(A) = 0$. (ii) Suppose B is a square matrix and $B^2 = B$. Show that B is diagonalizable.

(2) Let q be the minimal polynomial of a complex matrix A. Determine the minimal polynomial of A^*.

(3) Show that the matrix
$$M = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$
has no square root, that is there is no matrix N such that $N^2 = M$. (Hint: $M^2 = 0$.)

(4) Let $\mathcal{P}$ be the space of real polynomials of degree at most three. Let $D : \mathcal{P} \to \mathcal{P}$ be the linear map defined by
$$Df = f',$$
where f' denotes the derivative of f. Find the spectrum, minimal polynomial, characteristic polynomial of D. Is D diagonalizable?

(5) Write down the spectral decomposition in the form $\sum_j a_j P_j$, where P_j's are mutually orthogonal projections, for the following matrices
$$A = \begin{bmatrix} 3 & +i \\ -i & 3 \end{bmatrix}, B = \begin{bmatrix} 3 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}.$$

(6) Let N be a normal matrix. Show that a matrix M commutes with N and only if it commutes with N^*.

(7) Write down a matrix whose minimal polynomial is $q(x) = (x-2)(x-3)(x+4)$ and the characteristic polynomial is $p(x) = (x - 2)(x - 3)(x + 4)^2$.

(8) Write down polar and singular value decompositions for the following matrix:
$$C = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 4 & 0 \end{bmatrix}.$$

(9) Obtain simultaneous diagonalization for the following commuting matrices:
$$E = \begin{bmatrix} 5 & 2 & 0 \\ 2 & 5 & 0 \\ 0 & 0 & 3 \end{bmatrix}, F = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

(10) Write E, F of previous exercise as polynomials of a single matrix G.

(11) (Optional question) State and present a proof of Sylvester's law of inertia (Hint: See Wikipedia and other sources).